KALASIN STARCH PLANT - JET COOKER PROJECT

STANDARD INSTRUMENT HOOK-UP & TYPICAL ELECTRICAL LOOP DRAWINGS

Based on Equipment Packages in 11-instrument Manual/ & Master I/O Mapping Rev 3.6

DRAWING PACKAGE OBJECTIVES & ENGINEERING STANDARDS

Project Scope:	Jet Cooker Slurry & Flash Tank Instrumentation
Client / Owner:	INGREDION (THAILAND) CO., LTD.
Plant Location:	Kalasin Starch Manufacturing Plant, Thailand
System Integrator:	AEC INDUSTRIAL ENGINEERING CO., LTD.
Covered Packages:	23 Complete Equipment Types (CS-DS-01 to CS-DS-22)
Drawing Contents:	Piping Hook-Up, Loop Wiring Schematic & BOM
Electrical Standard:	Rockwell ControlLogix 1756 Architecture (24V DC / 4-20mA)
Hazardous Areas:	Ex ia / Ex d Classifications (937T Barriers & Glands)
CAD Standard:	1:1 Metric Millimeters (AutoCAD DXF R2010 Layered)
Piping Standard:	ASME B16.5 Cl.150/300 RF Flanges & NPT Threads
Document Status:	APPROVED ENGINEERING TEMPLATES (Rev 3.6)

DRAWING SYMBOLOGY & CAD LAYER CONVENTIONS

Process Piping:
Heavy solid line (0.7mm), showing process flow direction

Impulse Tubing:
Medium dashed line, 1/2" OD seamless SS316 tubing

Instrument Air:
Pneumatic air supply line, 6.0 bar dry filtered air

Analog Signal (AI/AO):
4-20mA HART loop, shielded twisted pair (Belden 8761)

Discrete Signal (DI/DO):
24V DC wetted contact or solenoid drive (Belden 9533)

IS Barrier (Ex ia):
Sky Blue block, Rockwell 937T Series Intrinsically Safe

Interposing Relay:
Orange block, Allen-Bradley 700-HL Series 24VDC Relay

Terminal Strip:
Green circle/box, Marshalling Panel CA1 (C#S##-X1)

Earth Grounding:
Protective Earth (PE) & Instrument Isolated Clean Earth (IE)

PREPARED BY	CHECKED BY	VERIFIED BY	APPROVED BY
AEC INDUSTRIAL ENG.	**AEC AUTOMATION LEAD**	**PROJECT ELECTRICAL PE**	**INGREDION PROJECT MGR.**
Lead Instrumentation Eng.	System Architect & PE	Licensed Professional Eng.	Client Automation Lead

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT			**AEC INDUSTRIAL ENGINEERING** SYSTEMS INTEGRATION & AUTOMATION
INSTRUMENT HOOK-UP & LOOP DRAWING DIRECTORY INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAM			**DWG NO:** KAL-JC-TYP-000 REV: 3.6 DATE: 2026-09-08
DRAWN: AEC-ENG CHECKED: LEAD-PE	**STATUS: APPROVED** SCALE: N.T.S.	**PROJECT NO:** R5THS00138-JC	**SHEET NO:** **01 / 25**